SERVICENOW SYSTEM ADMINISTRATOR · NAAN MUDHALVAN (SMARTBRIDGE)

Technology Stack - Native ServiceNow Configuration

Phase 2 · Requirement Analysis

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This document lists the technology components used to build the Script-Controlled ACL project. Every component is native to the ServiceNow Now Platform, running on a free Personal Developer Instance (dev443713.service-now.com); there is no external application, framework, or database.

Components and ServiceNow Technology

Component	ServiceNow Technology Used	Purpose in This Project
Platform	Now Platform - Personal Developer Instance (Australia / Zurich family release)	Hosts the custom table, security configuration, and evaluation engine; no external servers
Data model	Custom table u_institution_details (defined via sys_dictionary)	Stores the student Institution Details records and their columns
Choice field	sys_choice (Branch: ECE / EEE / CSE)	Supplies the field value that drives record-level access
Security engine	Access Control List (ACL) engine - sys_security_acl	Evaluates the read, create, write, and delete rules per record
ACL-to-role links	sys_security_acl_role	Binds each operation to its required role (bb1-bb4)
Scripting	Server-side GlideScript - gs.hasRole	READ ACL Advanced script returning true for admin and bb1, false otherwise
Roles and assignment	sys_user_role and sys_user_has_role	Defines bb1-bb4 and grants all four to the EEE User
Testing	Impersonation	Verifies access as EEE User, Abel Tuter, and System Administrator without separate logins
Admin control	security_admin role elevation	Required to create and edit the ACL rules
Scripting console	Background Scripts (System Definition > Scripts - Background)	Ad-hoc server-side checks and verification during the build